PHC 7091 Notes

Contents

1 Generalized Linear Model

1.1 Canonical GLM

Let $Y_1, \ldots, Y_n$ be independent random variables. A canonical GLM assumes that each Y_i follows an exponential dispersion family distribution with density

$$f(y_i \mid \theta_i, \phi) = \exp\left(\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right),$$

where

- θ_i is the *natural (canonical) parameter*,
- ϕ is the *dispersion parameter*,
- $a(\phi)$ is a known positive function,
- $b(\theta)$ is the cumulant (log-partition) function,
- $c(y, \phi)$ does not depend on θ_i.

Canonical link. The canonical link function satisfies

$$\theta_i = \eta_i = x_i^\top \beta,$$

where x_i is the covariate vector and β is the regression coefficient vector.

Mean and variance. The first two moments are

$$\mathbb{E}[Y_i] = \mu_i = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = a(\phi)\, b''(\theta_i).$$

Equivalently,

$$\mathrm{Var}(Y_i) = a(\phi)\, V(\mu_i), \quad V(\mu_i) = b''(\theta(\mu_i)).$$

Distribution	y	θ	$b(\theta)$	$a(\phi)$
Normal (μ, σ^2)	$y \in \mathbb{R}$	μ	$\dfrac{\theta^2}{2}$	σ^2
Poisson (λ)	$y \in \{0, 1, 2, \dots\}$	$\log \lambda$	e^θ	1
Binomial (n, p)	$y \in \{0, \dots, n\}$	$\log\left(\dfrac{p}{1-p}\right)$	$n \log(1 + e^\theta)$	1
Bernoulli (p)	$y \in \{0, 1\}$	$\log\left(\dfrac{p}{1-p}\right)$	$\log(1 + e^\theta)$	1
Gamma (μ, ϕ)	$y > 0$	$-\dfrac{1}{\mu}$	$-\log(-\theta)$	ϕ
Inverse Gaussian (μ, ϕ)	$y > 0$	$-\dfrac{1}{2\mu^2}$	$-\sqrt{-2\theta}$	ϕ

Example For the Gamma GLM, Gamma is conventionally parameterized as

$$Y_i \sim \text{Gamma}\left(\frac{1}{\phi}, \frac{1}{\phi\mu_i}\right),$$

where

$$\theta_i = -\frac{1}{\mu_i}, \qquad b(\theta_i) = -\log(-\theta_i), \qquad a(\phi) = \phi,$$

and

$$c(y_i, \phi) = \left(\frac{1}{\phi} - 1\right)\log y_i - \frac{\log\phi}{\phi} - \log\Gamma\left(\frac{1}{\phi}\right).$$

1.2 MLE θ for the Canonical Exponential Family

Consider independent observations $Y_1, \ldots, Y_n$ with density

$$f(y_i \mid \theta_i, \phi) = \exp\left(\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right),$$

where $a(\phi) > 0$ is known.

Log-likelihood. The log-likelihood (up to constants) is

$$\ell(\theta) = \frac{1}{a(\phi)}\sum_{i=1}^{n}\left(y_i\theta_i - b(\theta_i)\right).$$

Score function. The score function with respect to θ_i is

$$U(\theta_i) = \frac{\partial\ell}{\partial\theta_i} = \frac{1}{a(\phi)}\left(y_i - b'(\theta_i)\right).$$

Likelihood equation. The maximum likelihood estimator $\hat{\theta}_i$ satisfies

$$b'(\hat{\theta}_i) = y_i,$$

or equivalently,

$$\hat{\theta}_i = (b')^{-1}(y_i),$$

provided $b'(\theta)$ is invertible.

Existence and uniqueness. If $b(\theta)$ is strictly convex, then $b'(\theta)$ is strictly increasing, and the MLE $\hat{\theta}_i$ exists and is unique whenever y_i lies in the interior of the mean parameter space.

Observed information. The observed Fisher information is

$$-\frac{\partial^2\ell}{\partial\theta_i^2} = \frac{1}{a(\phi)}b''(\theta_i).$$

Fisher information. The Fisher information for θ_i is

$$\mathcal{I}(\theta_i) = \mathbb{E}\left[-\frac{\partial^2 \ell}{\partial \theta_i^2}\right] = \frac{1}{a(\phi)} b''(\theta_i).$$

Asymptotic distribution. Under standard regularity conditions,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{a(\phi)}{b''(\theta)}\right).$$

Invariance property. Since $\mu = b'(\theta)$, the MLE of the mean parameter is

$$\hat{\mu} = b'(\hat{\theta}).$$

Efficiency. The MLE $\hat{\theta}$ attains the Cramér–Rao lower bound:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{a(\phi)}{n\, b''(\theta)},$$

with equality asymptotically.

Newton update. Newton's method updates θ_i by

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \left[\frac{\partial^2 \ell}{\partial \theta_i^2}\right]^{-1} \frac{\partial \ell}{\partial \theta_i}.$$

Substituting the derivatives gives

$$\boxed{\theta_i^{(t+1)} = \theta_i^{(t)} + \frac{y_i - b'(\theta_i^{(t)})}{b''(\theta_i^{(t)})}}$$

Interpretation. The Newton step moves θ_i in proportion to

$$\frac{\text{residual}}{\text{curvature}} = \frac{y_i - \mu_i^{(t)}}{\mathrm{Var}(Y_i)/a(\phi)}, \quad \mu_i^{(t)} = b'(\theta_i^{(t)}).$$

Convergence properties.

- If $b(\theta)$ is strictly convex, then $b''(\theta) > 0$ and the update is well-defined.
- The log-likelihood is concave in θ, so Newton iterations converge to the unique MLE.
- Convergence is quadratic in a neighborhood of $\hat{\theta}_i$.

Connection to Fisher scoring. Since

$$\mathcal{I}(\theta_i) = -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \theta_i^2}\right] = \frac{1}{a(\phi)} b''(\theta_i),$$

Newton's method coincides with Fisher scoring for canonical exponential families.

Special case (single observation). For a single observation y,

$$\theta^{(t+1)} = \theta^{(t)} + \frac{y - b'(\theta^{(t)})}{b''(\theta^{(t)})},$$

which converges to $\hat{\theta} = (b')^{-1}(y)$.

1.3 IRLS with Mean μ_i and Link Function $g(\mu_i)$

Let $Y_1, \dots, Y_n$ be independent with density

$$f(y_i \mid \theta_i, \phi) = \exp\left(\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right),$$

where

$$\mu_i = \mathbb{E}[Y_i] = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = a(\phi)\, b''(\theta_i).$$

Link function. A generalized linear model assumes

$$\eta_i = g(\mu_i) = x_i^\top \beta,$$

where $g(\cdot)$ is a monotone link function.

Log-likelihood. Up to constants, the log-likelihood is

$$\ell(\beta) = \frac{1}{a(\phi)} \sum_{i=1}^{n} \left(y_i\theta_i - b(\theta_i)\right), \quad \theta_i = \theta(\mu_i).$$

Score function. Using the chain rule,

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^{n} \frac{\partial \ell}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta}.$$

Since

$$\frac{\partial \ell}{\partial \mu_i} = \frac{y_i - \mu_i}{a(\phi)\, b''(\theta_i)}, \qquad \frac{\partial \eta_i}{\partial \beta} = x_i,$$

we obtain

$$\nabla_\beta \ell(\beta) = \frac{1}{a(\phi)} \sum_{i=1}^{n} \frac{y_i - b'(\theta_i)}{b''(\theta_i)} \frac{1}{g'(\mu_i)} x_i = \frac{1}{a(\phi)} \sum_{i=1}^{n} \frac{y_i - \mu_i}{V(\mu_i)} \frac{1}{g'(\mu_i)} x_i = \frac{1}{a(\phi)} \sum_{i=1}^{n} \frac{y_i - \mu_i}{\mathrm{Var}(\mu_i)} \frac{1}{g'(\mu_i)} x_i$$

Variance function. Define the variance function

$$V(\mu_i) = b''(\theta_i), \quad \mathrm{Var}(Y_i) = a(\phi)\, V(\mu_i).$$

Working weights. Define the weights

$$w_i = \frac{1}{a(\phi)} \left(\frac{d\mu_i}{d\eta_i}\right)^2 \frac{1}{V(\mu_i)} = \frac{1}{g'(\mu_i)^2 \mathrm{Var}(Y_i)} \frac{1}{a(\phi)}$$

Observed Hessian. Differentiating again gives the Hessian matrix

$$\frac{\partial^2 l}{\partial \beta_k \partial \beta_j} = \frac{1}{a(\phi)} \frac{\partial \sum_{i=1}^n (y_i - \mu_i) \frac{1}{g'(\mu_i)V(\mu_i)} x_{ik}}{\partial \beta_j}$$

$$= \frac{1}{a(\phi)} \sum_{i=1}^n \frac{\partial (y_i - \mu_i)}{\partial \beta_j} \frac{1}{g'(\mu_i)V(\mu_i)} + (y_i - \mu_i) \frac{\partial \frac{1}{g'(\mu_i)V(\mu_i)}}{\partial \beta_j} x_{ik}$$

$$= \{\frac{1}{a(\phi)} \sum_{i=1}^n - \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_i} \frac{1}{g'(\mu_i)V(\mu_i)} - (y_i - \mu_i)(\frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_i} \frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{(V(\mu_i)g'(\mu_i))^2})\} x_{ik}$$

so with

$$\frac{\partial \mu_i}{\partial \eta_j} = \frac{1}{g'(\mu_i)}, \frac{\partial \eta_i}{\partial \beta_j} = x_{ij}$$

we have it as

$$= \frac{1}{a(\phi)} \sum_{i=1}^n \{- \frac{1}{g'(\mu_i)^2 V(\mu_i)} - (y_i - \mu_i)(\frac{1}{g'(\mu_i)} \frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{(V(\mu_i)g'(\mu_i))^2})\} x_{ik} x_{ij}$$

Matrix form. Define

$$W_i^{\mathrm{obs}} = \frac{1}{a(\phi)} \left[\frac{1}{V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i}\right)^2 - (y_i - \mu_i) \frac{d}{d\mu_i} \left(\frac{1}{V(\mu_i)} \frac{d\mu_i}{d\eta_i} \right) \frac{d\mu_i}{d\eta_i} \right],$$

and let

$$W^{\mathrm{obs}} = \mathrm{diag}(W_1^{\mathrm{obs}}, \ldots, W_n^{\mathrm{obs}}).$$

Then

$$\boxed{\nabla_\beta^2 \ell(\beta) = -X^\top W^{\mathrm{obs}} X}$$

1.4 IRLS for β estimation

Expected Hessian (Fisher information). Taking expectation with respect to Y_i removes the residual term:

$$\mathbb{E}[y_i - \mu_i] = 0.$$

Hence the Fisher information matrix is

$$\boxed{\mathcal{I}(\beta) = -\mathbb{E}[\nabla^2_\beta \ell(\beta)] = X^\top W X}$$

with

$$\boxed{W_i = \frac{1}{a(\phi)} \frac{1}{V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i}\right)^2}$$

Hessian (expected). The expected Hessian is

$$\mathcal{I}(\beta) = X^\top W X, \quad W = \mathrm{diag}(w_1, \ldots, w_n).$$

Newton / Fisher scoring update. The Fisher scoring step is

$$\beta^{(t+1)} = \beta^{(t)} + \left(X^\top W^{(t)} X\right)^{-1} X^\top W^{(t)} (y - \mu^{(t)}),$$

where all quantities are evaluated at $\beta^{(t)}$.

Working response. Define the working response

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) \left(\frac{d\eta_i}{d\mu_i}\right)_{\mu_i^{(t)}}.$$

Weighted least squares form. Then $\beta^{(t+1)}$ solves

$$\beta^{(t+1)} = \arg\min_\beta \sum_{i=1}^n w_i^{(t)} \left(z_i^{(t)} - x_i^\top \beta\right)^2.$$

IRLS algorithm.

1. Initialize $\beta^{(0)}$.
2. At iteration t, compute
$$\eta_i^{(t)} = x_i^\top \beta^{(t)}, \quad \mu_i^{(t)} = g^{-1}(\eta_i^{(t)}).$$
3. Compute
$$w_i^{(t)} = \frac{1}{a(\phi)} \left(\frac{d\mu_i}{d\eta_i}\right)^2 \frac{1}{V(\mu_i)}.$$
4. Form
$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) \left(\frac{d\eta_i}{d\mu_i}\right).$$
5. Update
$$\beta^{(t+1)} = \left(X^\top W^{(t)} X\right)^{-1} X^\top W^{(t)} z^{(t)}.$$

Canonical link simplification. If $g(\mu) = \theta$, then

$$\frac{d\mu}{d\eta} = V(\mu), \quad w_i = \frac{V(\mu_i)}{a(\phi)},$$

and IRLS reduces to Newton's method exactly.

1.5 Derivation of the IRLS Update via Taylor Expansion

Assume a generalized linear model with

$$\eta_i = g(\mu_i) = x_i^\top \beta, \qquad \mathrm{Var}(Y_i) = a(\phi)\, V(\mu_i), \qquad \mu_i = \mathbb{E}[Y_i].$$

Score function. The score for β is

$$U(\beta) = \nabla_\beta \ell(\beta) = \sum_{i=1}^n \frac{y_i - \mu_i}{a(\phi)\, V(\mu_i)} \frac{d\mu_i}{d\eta_i} x_i.$$

First-order Taylor expansion of μ_i. Expand $\mu_i(\beta)$ around the current iterate $\beta^{(t)}$:

$$\mu_i(\beta) \approx \mu_i^{(t)} + \left.\frac{d\mu_i}{d\eta_i}\right|_{\beta^{(t)}} x_i^\top (\beta - \beta^{(t)}).$$

Linearization of the score. Substitute the expansion into the score equation $U(\beta) = 0$:

$$0 \approx \sum_{i=1}^n \frac{1}{a(\phi)\, V(\mu_i^{(t)})} \frac{d\mu_i}{d\eta_i} \left[y_i - \mu_i^{(t)} - \frac{d\mu_i}{d\eta_i} x_i^\top (\beta - \beta^{(t)}) \right] x_i.$$

Rearranging terms. Split the sum:

$$\sum_{i=1}^n \frac{y_i - \mu_i^{(t)}}{a(\phi)\, V(\mu_i^{(t)})} \frac{d\mu_i}{d\eta_i} x_i = \sum_{i=1}^n \frac{1}{a(\phi)} \frac{1}{V(\mu_i^{(t)})} \left(\frac{d\mu_i}{d\eta_i}\right)^2 x_i x_i^\top (\beta - \beta^{(t)}).$$

Matrix form. Define the weights

$$w_i^{(t)} = \frac{1}{a(\phi)} \frac{1}{V(\mu_i^{(t)})} \left(\frac{d\mu_i}{d\eta_i}\right)^2, \qquad W^{(t)} = \mathrm{diag}(w_1^{(t)}, \ldots, w_n^{(t)}).$$

Then

$$X^\top W^{(t)} X (\beta - \beta^{(t)}) = X^\top W^{(t)} \left[\eta^{(t)} + (y - \mu^{(t)}) \left(\frac{d\eta}{d\mu}\right) - X\beta^{(t)} \right].$$

Working response. Define the working response

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) \left(\frac{d\eta_i}{d\mu_i} \right).$$

Normal equations. The update satisfies

$$X^\top W^{(t)} X \, \beta^{(t+1)} = X^\top W^{(t)} z^{(t)}.$$

IRLS update. Thus

$$\boxed{\beta^{(t+1)} = \left(X^\top W^{(t)} X \right)^{-1} X^\top W^{(t)} z^{(t)}}$$

Canonical link simplification. If $g(\mu) = \theta$, then

$$\frac{d\mu}{d\eta} = V(\mu), \qquad w_i^{(t)} = \frac{V(\mu_i^{(t)})}{a(\phi)}, \qquad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \mu_i^{(t)}}{V(\mu_i^{(t)})}.$$

1.6 Initial Values for IRLS

Consider the model with only the intercept, then estimate β_0 with $\hat{\beta}_0$, set initial value as

$$(0, 0, \ldots, \hat{\beta}_0)$$

Example for Poisson GLM:

Step 1: Likelihood

Assume

$$Y_i \sim \text{Poisson}(\mu_i), \qquad \log(\mu_i) = \beta_0.$$

The Poisson density is

$$P(Y_i = y_i) = \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!}.$$

The log-likelihood for n independent observations is

$$l(\beta_0) = \sum_{i=1}^{n} \left[y_i \log(\mu_i) - \mu_i - \log(y_i!) \right].$$

Since $\mu_i = e^{\beta_0}$ and $\log(\mu_i) = \beta_0$,

$$l(\beta_0) = \sum_{i=1}^{n} \left[y_i \beta_0 - e^{\beta_0} - \log(y_i!) \right].$$

Step 2: Score Equation

Differentiate the log-likelihood:
$$\frac{\partial l(\beta_0)}{\partial \beta_0} = \sum_{i=1}^{n} \left(y_i - e^{\beta_0} \right).$$
Set the score equation equal to zero:
$$\sum_{i=1}^{n} y_i - ne^{\beta_0} = 0.$$

Step 3: Maximum Likelihood Estimator

Solving for β_0,
$$e^{\hat{\beta}_0} = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}.$$
Therefore
$$\hat{\beta}_0 = \log(\bar{y}).$$

Step 4: Update
$$\mu^{(0)} = \bar{y} \quad \text{or} \quad \frac{\bar{y} + y}{2}.$$
Let
$$\eta^{(0)} = X\beta^{(0)} + \text{offset}.$$

Define the working response
$$z = \eta^{(0)} - \text{offset} + \Delta\big(y - \mu^{(0)}\big),$$
where
$$\Delta = \frac{d\eta}{d\mu}.$$
For the Poisson log-link,
$$\eta = \log(\mu), \qquad \Delta = \frac{1}{\mu}.$$
The weight matrix is
$$W_0 = \text{diag}(\mu^{(0)}).$$
The first IRLS update is then
$$\beta^{(1)} \leftarrow \left(X^\top W_0 X \right)^{-1} X^\top W_0 z.$$

1.7 Equivalence of IRLS forms

Using the working response
$$z^{(t)} = \eta^{(t)} + \left.\frac{d\eta}{d\mu}\right|_{\mu^{(t)}} \left(y - \mu^{(t)} \right), \qquad \eta^{(t)} = X\beta^{(t)},$$

we have

$$\begin{aligned}\beta^{(t+1)} &= \left(X^\top W^{(t)} X\right)^{-1} X^\top W^{(t)} z^{(t)} \\ &= \left(X^\top W^{(t)} X\right)^{-1} X^\top W^{(t)} \left[X\beta^{(t)} + D^{(t)}(y - \mu^{(t)})\right] \\ &= \beta^{(t)} + \left(X^\top W^{(t)} X\right)^{-1} X^\top W^{(t)} D^{(t)}(y - \mu^{(t)}),\end{aligned}$$

where

$$D^{(t)} = \operatorname{diag}\left(\left.\frac{d\eta_i}{d\mu_i}\right|_{\mu_i^{(t)}}\right).$$

Therefore,

$$\beta^{(t+1)} = \beta^{(t)} + \left(X^\top W^{(t)} X\right)^{-1} X^\top W^{(t)}(y - \mu^{(t)})$$

holds only when $D^{(t)} = I$, such as under the identity link, or when $D^{(t)}$ is absorbed into the definition of the weight matrix.

2 Residual Test

Hat Matrix Consider the linear regression model

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2 I_n).$$

The fitted values are

$$\hat{Y} = X\hat{\beta}, \qquad \hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Define the *hat matrix*

$$H = X(X^\top X)^{-1} X^\top.$$

Then

$$\hat{Y} = HY.$$

The diagonal element h_{ii} of H is called the *leverage* of observation i.

$$\mathrm{Var}(e) = \sigma^2 (I - H)$$

Ordinary Residual The residual for observation i is

$$e_i = y_i - \hat{\mu}_i,$$

where

$$\hat{\mu}_i = x_i^\top \hat{\beta}.$$

Standardized Residual Let $\hat{\gamma}$ be an estimator of σ (often $\hat{\sigma}$). The standardized residual is

$$r_i = \frac{y_i - \hat{\mu}_i}{\hat{\gamma}}.$$

$$\mathrm{Var}(e_i) = \sigma^2 (1 - h_{ii}).$$

If $y_i \sim \mathrm{Poisson}(\mu_i)$. The standardized residual for observation i is

$$r_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}.$$

Pearson Goodness-of-Fit Statistic Let o_i denote the observed count and e_i denote the expected count under the fitted model, for $i = 1, \ldots, n$.

The Pearson goodness-of-fit statistic is

$$X^2 = \sum_{i=1}^{n} \frac{(o_i - e_i)^2}{e_i}$$

Under correct model specification and for large n,

$$X^2 \approx \chi^2_{n-p},$$

where p is the number of estimated parameters in the model.

$$\boxed{X_P^2 = \sum_j \frac{(O_j - E_j)^2}{E_j}}$$

This form is used when the observations are **cell counts or category frequencies**, such as:

- multinomial goodness-of-fit tests;
- contingency tables;
- Poisson count models.

For independent Poisson cell counts,

$$O_j \sim \text{Poisson}(E_j),$$

so that

$$E(O_j) = E_j, \qquad \text{Var}(O_j) = E_j.$$

Therefore,

$$\frac{(O_j - E_j)^2}{E_j} = \frac{(O_j - E(O_j))^2}{\text{Var}(O_j)},$$

and hence

$$X_P^2 = \sum_j \frac{(O_j - E(O_j))^2}{\text{Var}(O_j)}.$$

For multinomial or contingency-table data, the cell counts are not independent because their total is fixed. However, a multinomial distribution can be obtained by conditioning independent Poisson counts on their total. Therefore, the same Pearson statistic is used.

For example, if

$$Y \sim \text{Binomial}(n, \pi),$$

the two cell counts are

$$O_1 = Y, \qquad O_2 = n - Y,$$

with expected counts

$$E_1 = n\pi, \qquad E_2 = n(1 - \pi).$$

Then

$$\begin{aligned} X_P^2 &= \frac{(Y - n\pi)^2}{n\pi} + \frac{[(n - Y) - n(1 - \pi)]^2}{n(1 - \pi)} \\ &= \frac{(Y - n\pi)^2}{n\pi(1 - \pi)} \\ &= \frac{(Y - E(Y))^2}{\text{Var}(Y)}. \end{aligned}$$

Thus,

$$\boxed{\begin{array}{ll} \displaystyle\sum_i \frac{(Y_i-\mu_i)^2}{\mathrm{Var}(Y_i)} & \text{is used for general model responses,} \\ \displaystyle\sum_j \frac{(O_j-E_j)^2}{E_j} & \text{is used for cell-count or frequency data.} \end{array}}$$

The denominator E_j is appropriate because cell counts are modeled directly as Poisson counts, or as multinomial counts obtained by conditioning Poisson counts on a fixed total.

The connection between fixed total Poisson and Multinomial model will be provided in 2.3

2.1 Testing Nested Models in GLMs

Let M_0 denote the model under the null hypothesis and M_1 the model under the alternative hypothesis, where M_0 is nested within M_1.

The procedure for testing nested models in a generalized linear model (GLM) is:

1. Specify the null model M_0 and the alternative model M_1.
2. Compute the goodness-of-fit statistics G_0 for M_0 and G_1 for M_1.
3. Form a test statistic such as
$$G_1 - G_0 \quad \text{or} \quad \frac{G_1}{G_0}.$$
4. Reject the null hypothesis based on the distribution of the chosen test statistic.

In GLMs, the difference in deviances between nested models is commonly used and asymptotically follows a χ^2 distribution.

The **Akaike Information Criterion (AIC)** is defined as

$$AIC = -2\ell + 2p,$$

where ℓ is the maximized log-likelihood and p is the number of parameters.

The **Bayesian Information Criterion (BIC)** is defined as

$$BIC = -2\ell + p\log(n),$$

where n is the sample size.

Models with smaller AIC or BIC values are preferred.

Scaled Deviance Let $Y_1, \dots, Y_n$ be independent observations from an exponential family distribution with log-likelihood

$$\ell(\boldsymbol{\mu}; \mathbf{y}) = \sum_{i=1}^{n} \frac{y_i \theta_i - b(\theta_i)}{\phi} + \sum_{i=1}^{n} c(y_i, \phi),$$

where θ_i is the canonical parameter corresponding to the mean μ_i, and ϕ is the dispersion parameter.

The scaled deviance is defined as

$$D(\mathbf{y}; \boldsymbol{\mu}) = 2 \left[\ell(\widehat{\boldsymbol{\mu}}^{\text{sat}}; \mathbf{y}) - \ell(\boldsymbol{\mu}; \mathbf{y}) \right],$$

where

- $\ell(\boldsymbol{\mu}; \mathbf{y})$ is the log-likelihood evaluated at the fitted mean vector $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)$;
- $\widehat{\boldsymbol{\mu}}^{\text{sat}}$ is the fitted mean vector under the saturated model, which **assigns a separate mean parameter to every observation**. When allowed by the mean parameter space, this gives

$$\widehat{\mu}_i^{\text{sat}} = y_i.$$

Thus, the deviance measures twice the difference between the log-likelihood of the saturated model and that of the fitted model. It quantifies the loss of fit resulting from imposing the structure of the fitted GLM rather than allowing each observation to have its own mean parameter.

2.2 Residual test examples

Pearson residual (Poisson case):

$$r_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$$

Pearson residual (Gamma case):

$$r_i = \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i}$$

Under the model assumptions,

$$\sum_{i=1}^{n} r_i^2 \sim \chi^2_{n-p}.$$

Deviance:

$$D(y, \hat{\mu}) = \sum_{i=1}^{n} 2\phi \left[\ell(y_i; y_i) - \ell(\hat{\mu}_i; y_i) \right].$$

Deviance residual:

$$r_i^d = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i},$$

where d_i is the individual contribution to the deviance.

The scaled deviance satisfies

$$\frac{D(y, \hat{\mu})}{\phi} \sim \chi^2_{n-p}.$$

Null Deviance.

The *null deviance* is obtained by fitting the **null model**, which contains only an intercept (no covariates). It is computed by comparing the saturated model with this intercept-only model.

Likelihood Ratio Test for Nested GLMs Let $M_0 \subset M_1$ be two nested GLMs with $p_0 < p_1$ parameters. Let $D(y, \hat{\mu}_0)$ and $D(y, \hat{\mu}_1)$ denote the deviances under M_0 and M_1.

The likelihood ratio test statistic is

$$G = \frac{D(y, \hat{\mu}_0) - D(y, \hat{\mu}_1)}{\phi} = 2(\ell(y, \hat{\mu}_{full}) - \ell(y, \hat{\mu}_{reduced})) \sim \chi^2_{p.full-p.reduced}$$

ϕ can be estimated from the full model

Example: Poisson Regression Assume the number of deaths Y follows a Poisson distribution

$$f(y; \mu) = \frac{\mu^y e^{-\mu}}{y!}.$$

The expected number of deaths depends on the population size M and a mortality rate that depends on covariates x. Thus we model

$$E(Y) = \mu = M\mathcal{X}(x'\beta),$$

where $\mathcal{X}(x'\beta)$ represents the mortality rate.

A common specification (Dobson) is

$$E(Y) = \mu = Me^{x'\beta}.$$

Using the log link,

$$g(\mu_i) = \log(\mu_i) = \log(M_i) + x_i'\beta,$$

where $\log(M_i)$ acts as an offset.

Detail at textbook 9.2, P198

$$E\left(Y_i\right) = \mu_i = n_i\theta_i$$

$$\theta_i = e^{\mathbf{x}_i^T\boldsymbol{\beta}}$$

The GLM model is

$$E\left(Y_i\right) = \mu_i = n_i e^{\mathbf{x}_i^T\boldsymbol{\beta}}; \quad Y_i \sim \operatorname{Po}\left(\mu_i\right)$$

Link:

$$\log\mu_i = \log n_i(\text{offset}) + \mathbf{x}_i^T\boldsymbol{\beta}$$

and we have rate ratio

$$RR = \frac{\mathrm{E}\left(Y_i \mid \text{ present }\right)}{\mathrm{E}\left(Y_i \mid \text{ absent }\right)} = e^{\beta_j}$$

All the other explanatory variables remain the same, for a continuous explanatory variable x_k, a one-unit increase will result in a multiplicative effect of e^{β_k} on the rate μ.

For a binary explanatory variable denoted by an indictor variable, $x_j = 0$ if the factor is absent and $x_j = 1$ if it is present.

$$\frac{b_j - \beta_j}{\text{s.e. } (b_j)} \sim N(0,1)$$

where $b_j = \hat{\beta}_j$ is the estimator. For Wald test, confidence intervals, score, LRT, etc. Fit

$$\widehat{Y}_i = \widehat{\mu}_i = n_i e^{\mathbf{x}_i^T \mathbf{b}}$$

Also denoted as e_i ($e_i = \hat{Y}_i$)

Pearson residuals

$$r_i = \frac{o_i - e_i}{\sqrt{e_i}} = \frac{Y_i - \hat{Y}_i}{\sqrt{\hat{Y}_i}}$$

o_i denotes the observed value of Y_i. The deviance is given by

$$D = 2\sum \left[o_i \log\left(o_i/e_i\right) - \left(o_i - e_i\right)\right]$$

However, for most models $\sum o_i = \sum e_i$, so the deviance simplifies to

$$D = 2\sum \left[o_i \log\left(o_i/e_i\right)\right].$$

With Tarlor expansion

$$o \log\left(\frac{o}{e}\right) = (o - e) + \frac{1}{2}\frac{(o-e)^2}{e} + \dots$$

we have

$$\begin{aligned} D &= 2\sum_{i=1}^{N}\left[\left(o_i - e_i\right) + \frac{1}{2}\frac{\left(o_i - e_i\right)^2}{e_i} - \left(o_i - e_i\right)\right] \\ &= \sum_{i=1}^{N}\frac{\left(o_i - e_i\right)^2}{e_i} = X^2 = \sum r_i^2. \end{aligned}$$

Both D and X^2 follows χ^2_{N-p}. where N is the number of independent observations and p is the number of estimated regression parameters, including the intercept. The offset is known and is therefore not counted as an estimated parameter.

Example 9.2.1, P201

2.3 Connection between fixed total Poisson and Multinomial model

Example: Multinomial Model If the only constraint is that the sum of the Y_i 's is n, then the following Multinomial distribution may be used

$$f(\mathbf{y}; \boldsymbol{\mu} \mid n) = n! \prod_{i=1}^{N} \theta_i^{y_i} / y_i!,$$

where $\sum_{i=1}^N \theta_i = 1$ and $\sum_{i=1}^N y_i = n$. In this case, $E(Y_i) = n\theta_i$.

Example: Poisson Model with Different rate Let $Y_1, \ldots, Y_T$ be independent random variables such that

$$Y_i \sim \text{Poisson}(\lambda_i), \qquad i = 1, \ldots, T.$$

The probability mass function of Y_i is

$$P(Y_i = y_i) = \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}.$$

Given observations $y_1, \ldots, y_T$, the likelihood function is

$$L(\lambda_1, \ldots, \lambda_T; y_1, \ldots, y_T) = \prod_{i=1}^T \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}.$$

that

$$\sum Y_i \sim \text{Poisson}(\sum \lambda_i)$$

The log-likelihood is

$$\ell(\lambda_1, \ldots, \lambda_T) = \sum_{i=1}^T \left(-\lambda_i + y_i \log \lambda_i - \log(y_i!)\right).$$

Poisson and Multinomial Conversion Let $Y_1, \ldots, Y_T$ be independent random variables with

$$Y_i \sim \text{Poisson}(\lambda_i), \qquad i = 1, \ldots, T.$$

Then

$$S = \sum_{i=1}^T Y_i \sim \text{Poisson}\left(\sum_{i=1}^T \lambda_i\right).$$

Moreover, conditional on $S = s$,

$$(Y_1, \ldots, Y_T) \mid S = s \sim \text{Multinomial}\left(s; \frac{\lambda_1}{\sum_{i=1}^T \lambda_i}, \ldots, \frac{\lambda_T}{\sum_{i=1}^T \lambda_i}\right).$$

Proof. Let $S = \sum_{i=1}^T Y_i$. Since the Y_i are independent,

$$P(Y_1 = y_1, \ldots, Y_T = y_T) = \prod_{i=1}^T \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}.$$

To compute the distribution of S, write

$$P(S = s) = \sum_{\substack{y_1 + \cdots + y_T = s \\ y_i \geq 0}} P(Y_1 = y_1, \ldots, Y_T = y_T).$$

Substituting the joint pmf gives

$$P(S=s) = \sum_{y_1+\cdots+y_T=s} \prod_{i=1}^{T} \frac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!}.$$

Factor out the terms not depending on $(y_1, \dots, y_T)$:

$$P(S=s) = e^{-\sum_{i=1}^{T}\lambda_i} \sum_{y_1+\cdots+y_T=s} \prod_{i=1}^{T} \frac{\lambda_i^{y_i}}{y_i!}.$$

Use the multinomial expansion

$$(\lambda_1 + \cdots + \lambda_T)^s = \sum_{y_1+\cdots+y_T=s} \frac{s!}{y_1!\cdots y_T!} \prod_{i=1}^{T} \lambda_i^{y_i}.$$

Hence

$$\sum_{y_1+\cdots+y_T=s} \prod_{i=1}^{T} \frac{\lambda_i^{y_i}}{y_i!} = \frac{(\sum_{i=1}^{T}\lambda_i)^s}{s!}.$$

Therefore

$$P(S=s) = e^{-\sum_{i=1}^{T}\lambda_i} \frac{(\sum_{i=1}^{T}\lambda_i)^s}{s!},$$

which is the pmf of

$$\text{Poisson}\left(\sum_{i=1}^{T}\lambda_i\right).$$

Now compute the conditional distribution. For $y_1 + \cdots + y_T = s$,

$$P(Y_1 = y_1, \dots, Y_T = y_T \mid S = s) = \frac{P(Y_1 = y_1, \dots, Y_T = y_T)}{P(S=s)}.$$

Substituting the expressions above and simplifying gives

$$P(Y_1 = y_1, \dots, Y_T = y_T \mid S = s) = \frac{s!}{y_1!\cdots y_T!} \prod_{i=1}^{T} \left(\frac{\lambda_i}{\sum_{j=1}^{T}\lambda_j}\right)^{y_i}.$$

This is the pmf of a multinomial distribution with parameters

$$s \quad \text{and} \quad p_i = \frac{\lambda_i}{\sum_{j=1}^{T}\lambda_j}.$$

□

Therefore, conditioning on $S = s$, since

$$(Y_1, \dots, Y_T) \mid S = s \sim \text{Multinomial}(s; p_1, \dots, p_T), \qquad p_i = \frac{\lambda_i}{\sum_{j=1}^{T}\lambda_j},$$

we have

$$E(Y_i \mid S = s) = sp_i,$$

$$\mathrm{Var}(Y_i \mid S = s) = sp_i(1 - p_i),$$

and for $i \neq j$,

$$\mathrm{Cov}(Y_i, Y_j \mid S = s) = -sp_ip_j.$$

Log-linear model is often used for associations.

Example 9.7.2 P215, 9.3.2 on P208

7.2, 7.3, 7.4

3 Binomial Model GLM

Binomial GLM Let $Y_1, \ldots, Y_N$ be independent random variables such that

$$Y_i \sim \text{Binomial}(\mu_i, \pi_i), \qquad i = 1, \ldots, N,$$

where μ_i is the number of trials and π_i is the success probability.

The generalized linear model specifies that the mean

$$E(Y_i) = \mu_i \pi_i$$

is linked to the covariates through

$$g(\pi_i) = x_i' \beta,$$

where $g(\cdot)$ is the link function.

The likelihood function is

$$L(\pi; y) = \prod_{i=1}^{N} \binom{\mu_i}{y_i} \pi_i^{y_i} (1 - \pi_i)^{\mu_i - y_i}.$$

The log-likelihood is

$$\ell(\pi; y) = \sum_{i=1}^{N} \left[y_i \log(\pi_i) + (\mu_i - y_i) \log(1 - \pi_i) + \log \binom{\mu_i}{y_i} \right].$$

The maximum likelihood estimator $\hat{\beta}$ maximizes $\ell(\pi; y)$ subject to

$$g(\pi_i) = x_i' \beta.$$

Table 7.1

Deviance for the Binomial GLM Let $Y_1, \ldots, Y_N$ be independent with

$$Y_i \sim \text{Binomial}(\mu_i, \pi_i), \qquad E(Y_i) = \mu_i \pi_i.$$

Let $\hat{\pi}_i$ denote the fitted probabilities from the GLM and $\hat{\mu}_i = \mu_i \hat{\pi}_i$ the fitted means.

The deviance is defined as

$$D(y; \hat{\mu}) = 2\big[l(y; y) - l(\hat{\mu}; y)\big].$$

For the binomial model this becomes

$$D(y; \hat{\mu}) = 2 \sum_{i=1}^{N} \left[y_i \log\left(\frac{y_i}{\hat{\mu}_i} \right) + (\mu_i - y_i) \log\left(\frac{\mu_i - y_i}{\mu_i - \hat{\mu}_i} \right) \right].$$

That is

$$D(y; \hat{\mu}) = 2 \sum_{i=1}^{N} \left[y_i \log\left(\frac{y_i}{\hat{\mu}_i} \right) \right].$$

that

$$D=2\sum o\log(o/e)$$

that o is the observation (y_i or $y_i-\mu_i$), e is the respective frequency (fitted values). $\widehat{y}_i=n_i\widehat{\pi}_i$ and $(n_i-\widehat{y}_i)=(n_i-n_i\widehat{\pi}_i)$ that

$$D\sim\chi^2_{N-p}$$

Suppose model 1 with p_1 parameters and model 2 with p_2 parameters $p_1<p_2$ that

$$\Delta D=D_1-D_2\sim\chi^2_{p_2-p_1}$$

This could be taken for examine if we prefer model 1 or 2. We can also use AIC

(7.4 P158)

(7.5, P162)

Binomial GLM Goodness of fit

$$S_w=\sum_{i=1}^{N}\frac{(y_i-n_i\pi_i)^2}{n_i\pi_i(1-\pi_i)}$$

since $E(Y_i)=n_i\pi_i$ and $\operatorname{Var}(Y_i)=n_i\pi_i(1-\pi_i)$. This is equivalent to minimizing the Pearson chi-squared statistic

$$X^2=\sum\frac{(o-e)^2}{e},$$

Proof.

$$\begin{aligned}X^2&=\sum_{i=1}^{N}\frac{(y_i-n_i\pi_i)^2}{n_i\pi_i}+\sum_{i=1}^{N}\frac{[(n_i-y_i)-n_i(1-\pi_i)]^2}{n_i(1-\pi_i)}\\&=\sum_{i=1}^{N}\frac{(y_i-n_i\pi_i)^2}{n_i\pi_i(1-\pi_i)}(1-\pi_i+\pi_i)=S_w.\end{aligned}$$

□

X^2 is estimated at

$$X^2=\sum_{i=1}^{N}\frac{(y_i-n_i\widehat{\pi}_i)^2}{n_i\widehat{\pi}_i(1-\widehat{\pi}_i)}$$

which is asymptotically equivalent to the deviance with taylor expansion

$$o\log(o/e)=(o-e)+\frac{1}{2}\frac{(o-e)^2}{e}+\dots$$

that gives

$$D = 2\sum_{i=1}^{N}\left\{(y_i - n_i\widehat{\pi}_i) + \frac{1}{2}\frac{(y_i - n_i\widehat{\pi}_i)^2}{n_i\widehat{\pi}_i} + [(n_i - y_i) - (n_i - n_i\widehat{\pi}_i)] \right.$$
$$\left. + \frac{1}{2}\frac{[(n_i - y_i) - (n_i - n_i\widehat{\pi}_i)]^2}{n_i - n_i\widehat{\pi}_i} + \dots \right\}$$
$$\cong \sum_{i=1}^{N}\frac{(y_i - n_i\widehat{\pi}_i)^2}{n_i\widehat{\pi}_i(1-\widehat{\pi}_i)} = X^2$$

Another goodness of fit testing is

$$C = 2[l(\widehat{\boldsymbol{\pi}}; \mathbf{y}) - l(\widetilde{\boldsymbol{\pi}}; \mathbf{y})]$$

for test statistics, where l is the log-likelihood function

$$C = 2\sum\left[y_i \log\left(\frac{\widehat{y}_i}{n_i\widetilde{\pi}_i}\right) + (n_i - y_i)\log\left(\frac{n_i - \widehat{y}_i}{n_i - n_i\widetilde{\pi}_i}\right)\right]$$

Under the minimal model $\widetilde{\pi} = (\Sigma y_i)/(\Sigma n_i)$. Let $\widehat{\pi}_i$ denote the estimated probability for Y_i under the model of interest (so the fitted value is $\widehat{y}_i = n_i\widehat{\pi}_i$).

This tests if all β entries are 0 or at least one of them is non-zero. the approximate sampling distribution for C is χ^2_{p-1} if all the p parameters except the intercept term are zero

(7.6 P166)

The Pearson, or chi-squared, residual is

$$X_k = \frac{(y_k - n_k\widehat{\pi}_k)}{\sqrt{n_k\widehat{\pi}_k(1-\widehat{\pi}_k)}} \quad , k = 1, \dots, m.$$

Deviance residuals can be defined similarly,

$$d_k = \operatorname{sign}(y_k - n_k\widehat{\pi}_k)\left\{2\left[y_k \log\left(\frac{y_k}{n_k\widehat{\pi}_k}\right) + (n_k - y_k)\log\left(\frac{n_k - y_k}{n_k - n_k\widehat{\pi}_k}\right)\right]\right\}^{1/2}$$

Both provides diagnotic information

- Residuals vs each explanatory variable (check for linearity)
- Plotted in the order of the measurements (check for serial correlation)
- Normal probability plots (Assume standardized residual should have approximately normal)

Example: Does response model We have the y_i as the success out of n_i trials, with the proportion of success as $p_i = y_i/n_i$. We have $E(p_i) = \pi_i$ and model $g(\pi_i) = x_i'\beta$ as the link function

$$f(y;\pi) = \binom{n}{np}\pi^{np}(1-\pi)^{n-np} = \exp\left\{\frac{p\log\frac{\pi}{1-\pi} + \log(1-\pi)}{1/n} + \log\binom{n}{np}\right\}.$$

with $\theta = \log(\pi/(1-\pi))$, it becomes

$$f(p;\pi) = \binom{n}{np} \exp\Big\{np\,\theta - n\log(1+e^{\theta})\Big\}\,,$$

$$\log f(p;\pi) = \log\binom{n}{np} + \frac{p\,\theta - \log(1+e^{\theta})}{1/n}$$

We have

$$\pi = \frac{e^{\theta}}{1+e^{\theta}} = E(p)$$

and

$$\mathrm{Var}(p_i) = \frac{a(\phi)}{n}\frac{e^{\theta_i}}{(1+e^{\theta_i})^2}.$$

with the link function

$$g(\pi_i) = \eta_i = \beta_1 + \beta_2 x_i$$

3.1 IRLS for Binomial GLM

Let $Y_i \sim \mathrm{Binomial}(n_i, \pi_i)$ and define the observed proportions

$$p_i = \frac{y_i}{n_i}, \qquad p = (p_1, \ldots, p_n)'.$$

Assume the GLM

$$g(\pi_i) = x_i'\beta, \qquad \eta = X\beta.$$

The MLE of β can be obtained using the **Iteratively Reweighted Least Squares (IRLS)** algorithm. At iteration m we solve the weighted least squares equation

$$(X'WX)^{(m-1)} b^{(m)} = X'Wz.$$

The components are defined as follows.

Weight matrix

$$W = \mathrm{diag}\bigg(\frac{1}{\mathrm{Var}(p_i)\,[g'(\pi_i)]^2}\bigg) = \mathrm{diag}\bigg(\frac{n_i}{\pi_i(1-\pi_i)\,[g'(\pi_i)]^2}\bigg)\,.$$

Adjusted response

$$z = Xb^{(m-1)} + \Delta(p-\pi) = \eta + \Delta(p-\pi),$$

evaluated at $\beta = b^{(m-1)}$.

Derivative matrix

$$\Delta = \mathrm{diag}\big(g'(\pi_i)\big)\,.$$

Initial estimate

$$\hat{b}^{(0)} = (X'X)^{-1}X'y.$$

The procedure iterates until convergence of $b^{(m)}$.

Example: IRLS for Binomial GLM

Let

$$Y_i \sim \text{Binomial}(n_i, \pi_i), \qquad p_i = \frac{y_i}{n_i}, \qquad p = (p_1, \dots, p_n)'.$$

Assume

$$\eta_i = \beta_1 + \beta_2 x_i, \qquad \pi_i = \phi(\eta_i).$$

The MLE of $\beta = (\beta_1, \beta_2)'$ can be obtained by the **Iteratively Reweighted Least Squares (IRLS)** algorithm. At iteration m we solve

$$(X'WX)^{(m-1)} b^{(m)} = X'Wz.$$

Weight matrix

$$W = \text{diag}\left(\frac{1}{\text{Var}(p_i)\,[\phi'(\eta_i)]^2}\right) = \text{diag}\left(\frac{n_i[\phi'(\eta_i)]^2}{\pi_i(1-\pi_i)}\right).$$

Adjusted response

$$z = Xb^{(m-1)} + \Delta(p - \pi) = \eta + \Delta(p - \pi),$$

where

$$\Delta = \text{diag}\left(\frac{1}{\phi'(\eta_i)}\right).$$

Design matrix

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}.$$

The iteration proceeds until $b^{(m)}$ converges.

Example: Logistic Regression Model

Let

$$Y_i \sim \text{Binomial}(n_i, \pi_i), \qquad p_i = \frac{y_i}{n_i}.$$

The logistic link function is

$$g(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i,$$

with linear predictor

$$\eta_i = \beta_1 + \beta_2 x_i.$$

Equivalently,

$$\pi_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}.$$

The maximum likelihood estimate of $\beta = (\beta_1, \beta_2)'$ can be obtained using the **Iteratively Reweighted Least Squares (IRLS)** algorithm, which at iteration m solves

$$(X'WX)^{(m-1)} b^{(m)} = X'Wz.$$

The matrices are

$$W = \mathrm{diag}(n_i \pi_i (1-\pi_i)),$$

$$\Delta = \mathrm{diag}(g'(\pi_i)) = \mathrm{diag}(\frac{1}{\pi_i(1-\pi_i)})$$

$$z = \eta + \frac{p-\pi}{\pi(1-\pi)},$$

and

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}.$$

The iteration proceeds until convergence of $b^{(m)}$.

Example: Binomial GLM with inverse log–log link

Let

$$Y_i \sim \mathrm{Binomial}(n_i, \pi_i), \qquad p_i = \frac{y_i}{n_i}.$$

The link function is

$$g(\pi_i) = \log\big(-\log(1-\pi_i)\big) = \eta_i, \qquad \eta_i = \beta_1 + \beta_2 x_i.$$

Hence

$$\pi_i = 1 - \exp\big(-e^{\eta_i}\big).$$

The MLE of $\beta = (\beta_1, \beta_2)'$ can be obtained by **Iteratively Reweighted Least Squares (IRLS)**

$$(X'WX)^{(m-1)} b^{(m)} = X'Wz.$$

Derivative of the inverse link

$$\phi'(\eta_i) = \frac{d\pi_i}{d\eta_i} = e^{\eta_i} \exp(-e^{\eta_i}).$$

Derivative of the link

$$g'(\pi_i) = \frac{1}{(1-\pi_i)\,[-\log(1-\pi_i)]}.$$

Weight matrix

$$W = \mathrm{diag}\left(\frac{1}{\mathrm{Var}(p_i)\,[g'(\pi_i)]^2}\right) = \mathrm{diag}\left(n_i \frac{(1-\pi_i)^2[\log(1-\pi_i)]^2}{\pi_i(1-\pi_i)}\right).$$

Derivative matrix

$$\Delta = \mathrm{diag}\big(g'(\pi_i)\big) = \mathrm{diag}\left(\frac{1}{(1-\pi_i)\,[-\log(1-\pi_i)]}\right).$$

Adjusted response

$$z = Xb^{(m-1)} + \Delta(p-\pi) = \eta + \Delta(p-\pi).$$

Example: Binomial GLM with log–log link

Let

$$Y_i \sim \text{Binomial}(n_i, \pi_i), \qquad p_i = \frac{y_i}{n_i}.$$

Assume the link

$$g(\pi_i) = \log(-\log \pi_i) = \eta_i, \qquad \eta_i = \beta_1 + \beta_2 x_i.$$

Hence the inverse link is

$$\pi_i = \exp(-e^{\eta_i}).$$

The MLE of β can be obtained by the **Iteratively Reweighted Least Squares (IRLS)** algorithm

$$(X'WX)^{(m-1)} b^{(m)} = X'Wz.$$

Variance of the proportion

$$\text{Var}(p_i) = \frac{\pi_i(1-\pi_i)}{n_i}.$$

Derivative of the link

$$g'(\pi_i) = -\frac{1}{\pi_i \log \pi_i}.$$

Weight matrix

$$W = \text{diag}\left(\frac{1}{\text{Var}(p_i)\,[g'(\pi_i)]^2}\right) = \text{diag}\left(\frac{n_i \pi_i (\log \pi_i)^2}{1-\pi_i}\right).$$

Derivative matrix

$$\Delta = \text{diag}\big(g'(\pi_i)\big) = \text{diag}\left(-\frac{1}{\pi_i \log \pi_i}\right).$$

Adjusted response

$$z = Xb^{(m-1)} + \Delta(p - \pi) = \eta + \Delta(p - \pi).$$

Example 7.3

4 Multinomial logistic model

4.1 Logistic Model with Binary Covariate

Suppose the response variable Y has J categories with probabilities

$$\pi_j = P(Y = j), \qquad j = 1, \dots, J,$$

and let $x \in \{0, 1\}$ be a binary covariate.

Taking category 1 as the baseline, the multinomial logit model is

$$\log\left(\frac{\pi_j}{\pi_1}\right) = \beta_{0j} + \beta_{1j}x, \qquad j = 2, \dots, J.$$

Interpretation for $x = 0$ and $x = 1$ When $x = 0$,

$$\log\left(\frac{\pi_{j0}}{\pi_{10}}\right) = \beta_{0j}.$$

When $x = 1$,

$$\log\left(\frac{\pi_{j1}}{\pi_{11}}\right) = \beta_{0j} + \beta_{1j}.$$

Here $\pi_{jk} = P(Y = j \mid x = k)$ for $k \in \{0, 1\}$.

Odds Ratio The odds ratio comparing $x = 1$ to $x = 0$ for category j relative to the baseline category 1 is

$$OR_j = \frac{\pi_{j1}/\pi_{11}}{\pi_{j0}/\pi_{10}}.$$

Using the model,

$$\log(OR_j) = \log\left(\frac{\pi_{j1}}{\pi_{11}}\right) - \log\left(\frac{\pi_{j0}}{\pi_{10}}\right) = (\beta_{0j} + \beta_{1j}) - \beta_{0j} = \beta_{1j}.$$

Therefore,

$$OR_j = e^{\beta_{1j}}.$$

Estimated Odds Ratio With estimated probabilities $\hat{\pi}_{jk}$,

$$\widehat{OR}_j = \frac{\hat{\pi}_{j1}/\hat{\pi}_{11}}{\hat{\pi}_{j0}/\hat{\pi}_{10}} = \exp(\hat{\beta}_{1j}).$$

The CI is

$$\exp\left(\hat{\beta}_{1j} \pm 1.96\sqrt{\widehat{\text{Var}}(\hat{\beta}_{1j})}\right)$$

which is equivalent to

$$\left(e^{\hat{\beta}_{1j} - 1.96SE\left(\hat{\beta}_{1j}\right)}, e^{\hat{\beta}_{1j} + 1.96SE\left(\hat{\beta}_{1j}\right)}\right)$$

4.2 Nominal and Ordinal Regression

Let

$$Y_i \in \{1, 2, \ldots, K\}, \qquad \Pr(Y_i = j \mid x_i) = \pi_{ij}, \qquad \sum_{j=1}^{K} \pi_{ij} = 1.$$

For nominal logistic regression, choose category K as the baseline. Then, for $j = 1, \ldots, K-1$,

$$\log\left(\frac{\pi_{ij}}{\pi_{iK}}\right) = x_i^\top \beta_j.$$

Equivalently,

$$\pi_{ij} = \frac{\exp(x_i^\top \beta_j)}{1 + \sum_{l=1}^{K-1} \exp(x_i^\top \beta_l)}, \qquad j = 1, \ldots, K-1,$$

and

$$\pi_{iK} = \frac{1}{1 + \sum_{l=1}^{K-1} \exp(x_i^\top \beta_l)}.$$

$$Y_i \in \{1, 2, \ldots, K\}, \qquad \Pr(Y_i \leq j \mid x_i) = \gamma_{ij}, \qquad j = 1, \ldots, K-1.$$

The ordinal logistic regression model (proportional odds model) is

$$\log\left(\frac{\Pr(Y_i \leq j \mid x_i)}{\Pr(Y_i > j \mid x_i)}\right) = \log\left(\frac{\gamma_{ij}}{1 - \gamma_{ij}}\right) = \alpha_j - x_i^\top \beta, \qquad j = 1, \ldots, K-1.$$

Equivalently,

$$\Pr(Y_i \leq j \mid x_i) = \gamma_{ij} = \frac{\exp(\alpha_j - x_i^\top \beta)}{1 + \exp(\alpha_j - x_i^\top \beta)}, \qquad j = 1, \ldots, K-1.$$

The category probabilities are

$$\Pr(Y_i = 1 \mid x_i) = \gamma_{i1},$$
$$\Pr(Y_i = j \mid x_i) = \gamma_{ij} - \gamma_{i,j-1}, \qquad j = 2, \ldots, K-1,$$
$$\Pr(Y_i = K \mid x_i) = 1 - \gamma_{i,K-1}.$$

Properties of the proportional odds model.

For an ordinal response $Y_i \in \{1, \ldots, K\}$, define

$$\gamma_{ij} = \Pr(Y_i \leq j \mid x_i), \qquad j = 1, \ldots, K-1.$$

The proportional odds model is

$$\log\left(\frac{\gamma_{ij}}{1 - \gamma_{ij}}\right) = \alpha_j - x_i^\top \beta.$$

Its main properties are:

1. It is a model for *ordered* categorical responses.
2. It models *cumulative logits*.
3. The slope vector β is the same for all cutpoints j (proportional odds assumption).
4. Only the intercepts α_j vary with j.
5. The cumulative logit curves are parallel in x_i.
6. The intercepts satisfy
$$\alpha_1 < \alpha_2 < \cdots < \alpha_{K-1}.$$
7. For a one-unit increase in x_r, the cumulative odds are multiplied by
$$\exp(-\beta_r),$$
uniformly over all cutpoints.
8. Equivalently, the odds of being in a higher category are multiplied by
$$\exp(\beta_r).$$
9. The category probabilities are recovered from the cumulative probabilities by
$$\Pr(Y_i = 1 \mid x_i) = \gamma_{i1},$$
$$\Pr(Y_i = j \mid x_i) = \gamma_{ij} - \gamma_{i,j-1}, \qquad j = 2, \dots, K-1,$$
$$\Pr(Y_i = K \mid x_i) = 1 - \gamma_{i,K-1}.$$
10. The model uses fewer parameters than nominal logistic regression.
11. When $K = 2$, it reduces to ordinary binary logistic regression.
12. It is appropriate only when the proportional odds assumption is reasonable.

Adjacent-categories logit model

$$\log\left(\frac{\pi_{ij}}{\pi_{i,j+1}}\right) = \alpha_j - x_i^\top \beta, \qquad j = 1, \dots, K-1.$$

Main properties:

1. It is for ordinal categorical responses.
2. It models *adjacent-category* odds, not cumulative odds.
3. The same slope vector β is used for all adjacent logits.
4. Only the intercepts α_j vary with j.
5. For a one-unit increase in x_r, the odds of category j versus $j+1$ are multiplied by
$$\exp(-\beta_r).$$
6. It has $(K-1) + p$ parameters.
7. It uses the ordering of the response categories.
8. When $K = 2$, it reduces to binary logistic regression.

Continuation-ratio logit model

$$\pi_{ij} = \Pr(Y_i = j \mid x_i), \qquad j = 1, \dots, K.$$

The continuation-ratio logit model is

$$\log\left(\frac{\Pr(Y_i = j \mid x_i)}{\Pr(Y_i > j \mid x_i)}\right) = \log\left(\frac{\pi_{ij}}{\pi_{i,j+1} + \cdots + \pi_{iK}}\right) = \alpha_j - x_i^\top \beta, \qquad j = 1, \dots, K-1.$$

Equivalently,

$$\log\left(\frac{\Pr(Y_i = j \mid Y_i \geq j,\ x_i)}{\Pr(Y_i > j \mid Y_i \geq j,\ x_i)}\right) = \alpha_j - x_i^\top \beta.$$

Main properties:

1. It is for ordinal categorical responses.
2. It models the odds of category j versus all higher categories.
3. It has a sequential interpretation.
4. The same slope vector β is used for all logits.
5. Only the intercepts α_j vary with j.
6. For a one-unit increase in x_r, the odds are multiplied by
$$\exp(-\beta_r).$$
7. It has $(K-1) + p$ parameters.
8. It reduces to binary logistic regression when $K = 2$.

Model can be compared with goodness of fit test

5 Quasi-likelihood for Generalized Linear Models

Let $Y_1, \ldots, Y_n$ be independent responses with

$$\mathbb{E}(Y_i) = \mu_i, \qquad \mathrm{Var}(Y_i) = \phi V(\mu_i),$$

where $V(\mu_i)$ is the variance function and $\phi > 0$ is the dispersion parameter.

In the quasi-likelihood approach, one does not assume a full probability model for Y_i. Instead, one seeks a function playing the role of a log-likelihood whose score with respect to μ_i has the same first two moments as the usual likelihood score in a correctly specified parametric model.

5.1 Quasi-score conditions

For each observation i, let $\ell_i(\mu_i; y_i)$ denote the contribution of the ith observation to a log-likelihood-type function. We require that its derivative with respect to μ_i satisfy the following three conditions:

$$\mathbb{E}\left(\frac{\partial \ell_i}{\partial \mu_i}\right) = 0, \tag{1}$$

$$\mathrm{Var}\left(\frac{\partial \ell_i}{\partial \mu_i}\right) = \frac{1}{\phi V(\mu_i)}, \tag{2}$$

$$-\mathbb{E}\left(\frac{\partial^2 \ell_i}{\partial \mu_i^2}\right) = \frac{1}{\phi V(\mu_i)}. \tag{3}$$

Thus the score is unbiased, and its variance equals the expected information.

Theorem 1 (Quasi-score identities)**.** Suppose

$$\mathbb{E}(Y_i) = \mu_i, \qquad \mathrm{Var}(Y_i) = \phi V(\mu_i).$$

Define

$$q_i(\mu_i; y_i) = \frac{y_i - \mu_i}{\phi V(\mu_i)}.$$

Then:

$$\begin{aligned}
\mathbb{E}\{q_i(\mu_i; Y_i)\} &= 0, \\
\mathrm{Var}\{q_i(\mu_i; Y_i)\} &= \frac{1}{\phi V(\mu_i)}, \\
-\mathbb{E}\left(\frac{\partial q_i(\mu_i; Y_i)}{\partial \mu_i}\right) &= \frac{1}{\phi V(\mu_i)}.
\end{aligned}$$

Hence q_i satisfies the quasi-likelihood score conditions.

Proof. First,

$$\mathbb{E}\{q_i(\mu_i; Y_i)\} = \mathbb{E}\left(\frac{Y_i - \mu_i}{\phi V(\mu_i)}\right) = \frac{\mathbb{E}(Y_i) - \mu_i}{\phi V(\mu_i)} = 0.$$

Second,

$$\mathrm{Var}\{q_i(\mu_i;Y_i)\} = \mathrm{Var}\left(\frac{Y_i-\mu_i}{\phi V(\mu_i)}\right) = \frac{\mathrm{Var}(Y_i)}{\phi^2 V(\mu_i)^2} = \frac{\phi V(\mu_i)}{\phi^2 V(\mu_i)^2} = \frac{1}{\phi V(\mu_i)}.$$

Finally, differentiating $q_i(\mu_i; y_i)$ with respect to μ_i gives

$$\frac{\partial q_i}{\partial \mu_i} = -\frac{1}{\phi V(\mu_i)} - \frac{(y_i-\mu_i)V'(\mu_i)}{\phi V(\mu_i)^2}.$$

Taking expectation,

$$\mathbb{E}\left(\frac{\partial q_i}{\partial \mu_i}\right) = -\frac{1}{\phi V(\mu_i)} - \frac{V'(\mu_i)}{\phi V(\mu_i)^2}\,\mathbb{E}(Y_i-\mu_i).$$

Since $\mathbb{E}(Y_i-\mu_i)=0$, this becomes

$$\mathbb{E}\left(\frac{\partial q_i}{\partial \mu_i}\right) = -\frac{1}{\phi V(\mu_i)}.$$

Therefore,

$$-\mathbb{E}\left(\frac{\partial q_i}{\partial \mu_i}\right) = \frac{1}{\phi V(\mu_i)}.$$

This proves the result. □

Since the quasi-score is defined by

$$\frac{\partial Q_i}{\partial \mu_i} = \frac{y_i-\mu_i}{\phi V(\mu_i)},$$

the associated quasi-likelihood contribution from the ith observation is obtained by integration:

$$Q_i(\mu_i;y_i) = \int_{y_i}^{\mu_i} \frac{y_i-t}{\phi V(t)}\,dt,$$

where t is a dummy variable of integration.

Equivalently,

$$Q_i(\mu_i;y_i) = -\int_{\mu_i}^{y_i} \frac{y_i-t}{\phi V(t)}\,dt.$$

By construction,

$$\frac{\partial Q_i(\mu_i;y_i)}{\partial \mu_i} = \frac{y_i-\mu_i}{\phi V(\mu_i)}.$$

The overall quasi-likelihood is then

$$Q(\mu;y) = \sum_{i=1}^{n} Q_i(\mu_i;y_i).$$

Example: normal case

Take

$$\phi = \sigma^2 = 1, \qquad b(\theta(\mu_i)) = \frac{\theta(\mu_i)^2}{2}.$$

Then for the normal exponential family,

$$\mu_i = b'(\theta_i) = \theta_i, \qquad V(\mu_i) = b''(\theta_i) = 1.$$

Hence

$$\frac{\partial Q_i}{\partial \mu_i} = \frac{y_i - \mu_i}{\phi V(\mu_i)} = y_i - \mu_i.$$

Integrating,

$$Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} (y_i - t)\, dt = -\frac{(y_i - \mu_i)^2}{2}.$$

Thus the total quasi-likelihood is

$$Q(\mu) = \sum_{i=1}^{n} Q_i = -\frac{1}{2}\sum_{i=1}^{n}(y_i - \mu_i)^2.$$

If $\mu_i = \theta$ for all i, then

$$Q(\theta) = -\frac{1}{2}\sum_{i=1}^{n}(y_i - \theta)^2.$$

Maximizing $Q(\theta)$ is equivalent to minimizing

$$\sum_{i=1}^{n}(y_i - \theta)^2.$$

So

$$\hat{\theta} = \bar{y}.$$

It is consistent but less efficient

$$\mathrm{Var}(\theta) \geq I(\theta)^{-1}$$

5.2 Quasi-deviance Computation

The quasi-deviance is defined by

$$D(y, \mu) = 2\sum_{i=1}^{n}\{Q_i(y_i; y_i) - Q_i(\mu_i; y_i)\}.$$

Equivalently, using the integral form of the quasi-likelihood,

$$D(y, \mu) = 2\sum_{i=1}^{n}\int_{\mu_i}^{y_i} \frac{y_i - t}{\phi V(t)}\, dt.$$

In the normal case with

$$\phi = 1, \qquad V(\mu_i) = 1,$$

we have

$$Q_i(\mu_i; y_i) = -\frac{(y_i - \mu_i)^2}{2}, \qquad Q_i(y_i; y_i) = 0.$$

Hence

$$D(y, \mu) = 2\sum_{i=1}^{n}\left(0 + \frac{(y_i - \mu_i)^2}{2}\right) = \sum_{i=1}^{n}(y_i - \mu_i)^2.$$

So for the Gaussian model, the quasi-deviance is just the residual sum of squares.

5.3 β estimation in a GLM via quasi-likelihood

In a GLM, let

$$g(\mu_i) = \eta_i = x_i^T\beta,$$

where $g(\cdot)$ is the link function. The quasi-likelihood estimator $\hat{\beta}$ is obtained by solving

$$\frac{\partial Q}{\partial \beta} = \sum_{i=1}^{n}\frac{\partial Q(\mu_i, y_i)}{\partial \mu_i}\frac{\partial \mu_i}{\partial \beta} = 0.$$

Since

$$\frac{\partial Q(\mu_i, y_i)}{\partial \mu_i} = \frac{y_i - \mu_i}{\phi V(\mu_i)},$$

we get the estimating equation

$$\sum_{i=1}^{n}\frac{y_i - \mu_i}{\phi V(\mu_i)}\frac{\partial \mu_i}{\partial \beta} = 0.$$

Now

$$\frac{\partial \mu_i}{\partial \beta} = \frac{d\mu_i}{d\eta_i}\frac{\partial \eta_i}{\partial \beta} = \frac{d\mu_i}{d\eta_i}x_i,$$

so the quasi-score for β is

$$U(\beta) = \sum_{i=1}^{n}\frac{(y_i - \mu_i)}{\phi V(\mu_i)}\frac{1}{g'(\mu_i)}x_i = 0.$$

In matrix form, this is

$$U(\beta) = \frac{D^T V^{-1}(y - \mu)}{\sigma^2} = 0,$$

where

$$V = diag(V(\mu_1), \ldots, V(\mu_n)), \quad D = diag(\frac{\partial \mu_1}{\partial \beta}, \ldots, (\frac{\partial \mu_n}{\partial \beta})$$

Thus $\hat{\beta}$ is the solution of the quasi-score equation

$$\sum_{i=1}^{n}\frac{(y_i - \mu_i)}{V(\mu_i)}\frac{d\mu_i}{d\eta_i}x_i = 0,$$

up to the constant factor $1/\phi$.

5.4 Newton–Raphson to solve Quasi-Score GLM

Write the quasi-score as

$$U(\beta) = \frac{D^T V^{-1}(y-\mu)}{\sigma^2} = 0,$$

where

$$V = \operatorname{diag}\bigl(V(\mu_1), \ldots, V(\mu_n)\bigr),$$

and

$$D = \frac{\partial \mu}{\partial \beta^T} = \begin{pmatrix} \frac{\partial \mu_1}{\partial \beta_1} & \cdots & \frac{\partial \mu_1}{\partial \beta_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial \mu_n}{\partial \beta_1} & \cdots & \frac{\partial \mu_n}{\partial \beta_p} \end{pmatrix}.$$

Thus D is the Jacobian matrix of $\mu = (\mu_1, \ldots, \mu_n)^T$ with respect to β.

Newton–Raphson updates β by

$$\beta^{(m+1)} = \beta^{(m)} - \left[\frac{\partial U(\beta)}{\partial \beta^T}\right]^{-1}_{\beta=\beta^{(m)}} U(\beta^{(m)}).$$

Using the usual Fisher scoring approximation,

$$-\frac{\partial U(\beta)}{\partial \beta^T} \approx \frac{D^T V^{-1} D}{\sigma^2},$$

so the update becomes

$$\beta^{(m+1)} = \beta^{(m)} + (D^T V^{-1} D)^{-1} D^T V^{-1}(y-\mu).$$

Equivalently,

$$\Delta\beta = (D^T V^{-1} D)^{-1} D^T V^{-1}(y-\mu), \qquad \beta^{(m+1)} = \beta^{(m)} + \Delta\beta.$$

This is the Newton–Raphson/Fisher scoring step for estimating β in the GLM quasi-likelihood framework.

$$U(\beta) = \frac{D^T V^{-1}(y-\mu)}{\sigma^2}.$$

Since

$$\operatorname{Cov}(y) = \sigma^2 V,$$

we have

$$\operatorname{Cov}\{U(\beta)\} = \frac{1}{\sigma^4} D^T V^{-1} \operatorname{Cov}(y)\, V^{-1} D = \frac{1}{\sigma^4} D^T V^{-1} (\sigma^2 V) V^{-1} D.$$

Therefore,

$$\operatorname{Cov}\{U(\beta)\} = \frac{1}{\sigma^2} D^T V^{-1} D.$$

6 Generalized Estimating Equations

For clustered/longitudinal data, let there be n independent subjects (clusters), where subject i has m_i repeated measurements.

The response vector for subject i is

$$Y_i = \begin{pmatrix} Y_{i1} \\ Y_{i2} \\ \vdots \\ Y_{im_i} \end{pmatrix}, \qquad i = 1, \ldots, n.$$

The corresponding covariate matrix is

$$X_i = \begin{pmatrix} X_{i1}^\top \\ X_{i2}^\top \\ \vdots \\ X_{im_i}^\top \end{pmatrix},$$

where each X_{ij} is a $p \times 1$ covariate vector, so that

$$X_i \in \mathbb{R}^{m_i \times p}, \qquad Y_i \in \mathbb{R}^{m_i}.$$

The marginal mean vector is

$$\mu_i = \mathbb{E}(Y_i \mid X_i) = \begin{pmatrix} \mu_{i1} \\ \mu_{i2} \\ \vdots \\ \mu_{im_i} \end{pmatrix}.$$

The GEE marginal model specifies

$$g(\mu_{ij}) = \eta_{ij} = X_{ij}^\top \beta, \qquad j = 1, \ldots, m_i,$$

or equivalently, in vector form,

$$g(\mu_i) = \begin{pmatrix} g(\mu_{i1}) \\ g(\mu_{i2}) \\ \vdots \\ g(\mu_{im_i}) \end{pmatrix} = X_i \beta.$$

The marginal variance is written as

$$\operatorname{Var}(Y_i \mid X_i) = V_i = A_i^{1/2} R_i(\alpha) A_i^{1/2},$$

where

$$A_i = \operatorname{diag}\big(\phi V(\mu_{i1}), \ldots, \phi V(\mu_{im_i})\big)$$

contains the marginal variances, and $R_i(\alpha)$ is the working correlation matrix for the repeated measurements within subject i.

For the marginal model

$$g(\mu_{ij}) = X_{ij}^\top \beta, \qquad \mu_i = \mathbb{E}(Y_i \mid X_i),$$

the generalized estimating equation (GEE) for β is

$$U(\beta) = \sum_{i=1}^{n} D_i^\top V_i^{-1}(Y_i - \mu_i) = 0,$$

where

$$D_i = \frac{\partial \mu_i}{\partial \beta^\top}, \qquad V_i = \mathrm{Var}(Y_i \mid X_i).$$

Using the GEE working covariance structure,

$$V_i = A_i^{1/2} R_i(\alpha) A_i^{1/2},$$

with

$$A_i = \mathrm{diag}\big(\phi V(\mu_{i1}), \dots, \phi V(\mu_{im_i})\big),$$

we get

$$\sum_{i=1}^{n} D_i^\top A_i^{-1/2} R_i(\alpha)^{-1} A_i^{-1/2}(Y_i - \mu_i) = 0.$$

This is the extension of the GLM quasi-score equation

$$\sum_{i=1}^{n} D_i^\top V_i^{-1}(Y_i - \mu_i) = 0$$

from independent data to clustered/longitudinal data by incorporating the within-cluster working correlation $R_i(\alpha)$.

6.1 Solving GEE

A common two-step iterative algorithm for solving the GEE is:

$$U(\beta) = \sum_{i=1}^{n} D_i^\top V_i^{-1}(Y_i - \mu_i) = 0, \qquad V_i = A_i^{1/2} R_i(\alpha) A_i^{1/2}.$$

Step 1: Update β given α, ϕ.

Given current estimates $\alpha^{(t)}, \phi^{(t)}$, form

$$V_i^{(t)} = A_i^{(t)1/2} R_i\left(\alpha^{(t)}\right) A_i^{(t)1/2},$$

and solve

$$\sum_{i=1}^{n} D_i^\top \big(V_i^{(t)}\big)^{-1}(Y_i - \mu_i) = 0$$

for β, usually by Fisher scoring / Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \left[\sum_{i=1}^{n} D_i^\top \big(V_i^{(t)}\big)^{-1} D_i\right]^{-1} \sum_{i=1}^{n} D_i^\top \big(V_i^{(t)}\big)^{-1}(Y_i - \mu_i).$$

Step 2: Update α and ϕ given β.

Using the updated $\beta^{(t+1)}$, compute fitted means

$$\mu_{ij}^{(t+1)} = g^{-1}(X_{ij}^\top \beta^{(t+1)}),$$

then update the nuisance parameters:

- estimate $\phi^{(t+1)}$ from Pearson residuals;
- estimate $\alpha^{(t+1)}$ from the within-cluster residual correlations.

Repeat Steps 1 and 2 until convergence of β (and usually also α, ϕ).

GEE estimation consistency

Under regularity conditions, the GEE estimator $\hat{\beta}$ is consistent and asymptotically normal:

$$\hat{\beta} \xrightarrow{p} \beta_0,$$

and

$$\sqrt{n}\,(\hat{\beta} - \beta_0) \xrightarrow{d} N\big(0,\ A^{-1}BA^{-1}\big)\,,$$

where

$$A = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} D_i^\top V_i^{-1} D_i,$$

and

$$B = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} D_i^\top V_i^{-1} \mathrm{Cov}(Y_i) V_i^{-1} D_i.$$

Thus the asymptotic covariance matrix is the *sandwich form*

$$\mathrm{Var}(\hat{\beta}) \approx \frac{1}{n} A^{-1} B A^{-1}.$$

If the working correlation is correctly specified, then $B = A$, so

$$\sqrt{n}\,(\hat{\beta} - \beta_0) \xrightarrow{d} N\big(0,\ A^{-1}\big)\,.$$

6.2 Covariance Matrix Estimation

A consistent estimator of the asymptotic covariance is the robust sandwich estimator. That can be estimated with:

$$\widehat{\mathrm{Var}}(\hat{\beta}) = \left(\sum_{i=1}^{n} D_i^\top V_i^{-1} D_i\right)^{-1} \left(\sum_{i=1}^{n} D_i^\top V_i^{-1} (Y_i - \mu_i)(Y_i - \mu_i)^\top V_i^{-1} D_i\right) \left(\sum_{i=1}^{n} D_i^\top V_i^{-1} D_i\right)^{-1}.$$

For the *compound symmetry* (exchangeable) working correlation structure, the within-cluster correlation matrix is

$$R_i(\alpha) = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{m_i \times m_i}.$$

Thus every pair of distinct observations in the same cluster has the same correlation:

$$\mathrm{Corr}(Y_{ij}, Y_{ik}) = \alpha, \qquad j \neq k.$$

The corresponding working covariance matrix is

$$V_i = A_i^{1/2} R_i(\alpha) A_i^{1/2}.$$

If all marginal variances are equal to σ^2, then

$$V_i = \sigma^2 R_i(\alpha).$$

Under the compound symmetry (exchangeable) working correlation structure, define the Pearson residual by

$$r_{ij} = \frac{Y_{ij} - \hat{\mu}_{ij}}{\sqrt{\hat{\phi}\, V(\hat{\mu}_{ij})}}.$$

Then the common correlation parameter α is typically estimated by the average of the within-cluster Pearson residual cross-products:

$$\hat{\alpha} = \frac{\sum_{i=1}^{n} \sum_{j<k} r_{ij} r_{ik}}{\sum_{i=1}^{n} \binom{m_i}{2}}.$$

Equivalently, writing it in terms of Y_{ij} and $\hat{\mu}_{ij}$,

$$\hat{\alpha} = \frac{\sum_{i=1}^{n} \sum_{j<k} \frac{(Y_{ij} - \hat{\mu}_{ij})(Y_{ik} - \hat{\mu}_{ik})}{\sqrt{\hat{\phi}\, V(\hat{\mu}_{ij})}\sqrt{\hat{\phi}\, V(\hat{\mu}_{ik})}}}{\sum_{i=1}^{n} \binom{m_i}{2}}.$$

So for exchangeable correlation, $\hat{\alpha}$ is just the average empirical Pearson correlation over all pairs within the same cluster.

Example: Gaussian Case

In the Gaussian case, suppose

$$Y_i \sim N(\mu_i, V_i), \qquad \mu_i = X_i \beta,$$

with clusters independent, i.e.

$$\operatorname{Cov}(Y_i, Y_{i'}) = 0, \qquad i \neq i'.$$

If each V_i is known, then the GEE becomes

$$\sum_{i=1}^{n} D_i^\top V_i^{-1}(Y_i - \mu_i) = 0.$$

Since

$$\mu_i = X_i\beta, \qquad D_i = \frac{\partial \mu_i}{\partial \beta^\top} = X_i,$$

the estimating equation is

$$\sum_{i=1}^{n} X_i^\top V_i^{-1}(Y_i - X_i\beta) = 0.$$

Solving for β gives the generalized least squares estimator

$$\hat{\beta} = \left(\sum_{i=1}^{n} X_i^\top V_i^{-1} X_i\right)^{-1} \left(\sum_{i=1}^{n} X_i^\top V_i^{-1} Y_i\right).$$

Equivalently, stacking all clusters,

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad X = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}, \qquad V = \operatorname{blockdiag}(V_1, \dots, V_n),$$

we have

$$\hat{\beta} = (X^\top V^{-1} X)^{-1} X^\top V^{-1} Y.$$

Its covariance matrix is

$$\operatorname{Var}(\hat{\beta}) = (X^\top V^{-1} X)^{-1}.$$

If the covariance matrices V_i are unknown, a common approach is *iterative feasible generalized least squares.*

Step 1: Initial estimate of β. Start with an initial working covariance, for example

$$V_i^{(0)} = I_{m_i},$$

and compute the OLS estimator

$$\beta^{(0)} = \left(\sum_{i=1}^{n} X_i^\top X_i\right)^{-1} \left(\sum_{i=1}^{n} X_i^\top Y_i\right).$$

Step 2: Estimate V_i from residuals. Using the current estimate $\beta^{(t)}$, form residuals

$$e_i^{(t)} = Y_i - X_i\beta^{(t)}.$$

Then estimate the covariance matrix V_i by a parametric working model

$$V_i = V_i(\theta),$$

with parameter θ estimated from the residuals $e_i^{(t)}$. For example, estimate variance and correlation parameters by moment methods or maximum likelihood.

Step 3: Update β. Given the updated covariance estimate $\hat{V}_i^{(t)}$, solve

$$\sum_{i=1}^{n} X_i^\top \big(\hat{V}_i^{(t)}\big)^{-1}(Y_i - X_i\beta) = 0,$$

so that

$$\beta^{(t+1)} = \left(\sum_{i=1}^{n} X_i^\top \big(\hat{V}_i^{(t)}\big)^{-1} X_i\right)^{-1} \left(\sum_{i=1}^{n} X_i^\top \big(\hat{V}_i^{(t)}\big)^{-1} Y_i\right).$$

Step 4: Iterate. Repeat Steps 2 and 3 until convergence of β and the covariance parameters. Thus the iterative scheme is

$$\beta^{(t)} \longrightarrow e_i^{(t)} = Y_i - X_i\beta^{(t)} \longrightarrow \hat{V}_i^{(t)} \longrightarrow \beta^{(t+1)}.$$

This is the usual *feasible GLS / iterative GEE* procedure in the Gaussian case.

The choice of V estimation

If $\hat{V}$ is substituted for V in the estimation of

$$\mathrm{Var}(\hat{\beta}) = (X^\top V^{-1} X)^{-1},$$

then the resulting estimator

$$(X^\top \hat{V}^{-1} X)^{-1}$$

usually underestimates the true variability of $\hat{\beta}$, because it treats the estimated covariance matrix $\hat{V}$ as if it were known and therefore ignores the additional variability from estimating V.

When V_i is replaced by $\hat{V}_i$, this plug-in variance formula does not account for the uncertainty in estimating the covariance parameters. As a result, it is typically too small.

A more reliable large-sample variance estimator is the sandwich estimator:

$$\widehat{\mathrm{Var}}(\hat{\beta}) = A^{-1} B A^{-1},$$

where

$$A = \sum_{i=1}^{n} X_i^\top \hat{V}_i^{-1} X_i,$$

and

$$B = \sum_{i=1}^{n} X_i^\top \hat{V}_i^{-1} (Y_i - X_i\hat{\beta})(Y_i - X_i\hat{\beta})^\top \hat{V}_i^{-1} X_i.$$

Thus,

$$\widehat{\mathrm{Var}}(\hat{\beta}) = \left(\sum_{i=1}^{n} X_i^\top \hat{V}_i^{-1} X_i\right)^{-1} \left(\sum_{i=1}^{n} X_i^\top \hat{V}_i^{-1} (Y_i - X_i\hat{\beta})(Y_i - X_i\hat{\beta})^\top \hat{V}_i^{-1} X_i\right) \left(\sum_{i=1}^{n} X_i^\top \hat{V}_i^{-1} X_i\right)^{-1}.$$

This estimator remains consistent even if the working covariance structure is misspecified, and it corrects for the downward bias of the naive plug-in estimator. It is consistent and robust to the misspecification of V.

Small-Sample Variance Estimation in GEE

Although the sandwich estimator is consistent as the number of independent clusters K tends to infinity, it tends to underestimate the true variance when the number of clusters is small. In particular, when $K \leq 30$, the resulting standard errors may be too small and Wald tests may have inflated Type I error rates.

A cluster-level jackknife variance estimator can be used to reduce this small-sample bias. Let

$$\widehat{\boldsymbol{\beta}}_{-i}$$

denote the GEE estimator obtained after deleting the entire i-th cluster. The delete-one-cluster jackknife variance estimator is

$$\widehat{\mathrm{Var}}_{\mathrm{JK}}\left(\widehat{\boldsymbol{\beta}}\right) = \frac{K-1}{K} \sum_{i=1}^{K} \left(\widehat{\boldsymbol{\beta}}_{-i} - \overline{\boldsymbol{\beta}}_{(-)}\right) \left(\widehat{\boldsymbol{\beta}}_{-i} - \overline{\boldsymbol{\beta}}_{(-)}\right)^T,$$

where

$$\overline{\boldsymbol{\beta}}_{(-)} = \frac{1}{K} \sum_{i=1}^{K} \widehat{\boldsymbol{\beta}}_{-i}.$$

An alternative finite-sample version, sometimes used in GEE applications, centers the deleted-cluster estimates at the full-sample estimate and applies a degrees-of-freedom correction:

$$\widehat{\mathrm{Var}}_{\mathrm{JK}}\left(\widehat{\boldsymbol{\beta}}\right) = \frac{K-p}{K} \sum_{i=1}^{K} \left(\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}}\right) \left(\widehat{\boldsymbol{\beta}}_{-i} - \widehat{\boldsymbol{\beta}}\right)^T,$$

where p is the number of estimated regression parameters. The exact scaling convention may differ across references and software packages.

The jackknife standard error for the j-th regression coefficient is

$$\mathrm{SE}_{\mathrm{JK}}\left(\widehat{\beta}_j\right) = \sqrt{\left[\widehat{\mathrm{Var}}_{\mathrm{JK}}\left(\widehat{\boldsymbol{\beta}}\right)\right]_{jj}}.$$

For a small number of clusters, inference is commonly based on a t-distribution rather than the standard normal distribution:

$$T_j = \frac{\widehat{\beta}_j - \beta_{j,0}}{\mathrm{SE}_{\mathrm{JK}}\left(\widehat{\beta}_j\right)} \mathrel{\dot\sim} t_{K-p}.$$

An approximate $100(1-\alpha)\%$ confidence interval is therefore

$$\widehat{\beta}_j \pm t_{K-p,\,1-\alpha/2}\,\mathrm{SE}_{\mathrm{JK}}\left(\widehat{\beta}_j\right).$$

Thus, the ordinary sandwich estimator is generally appropriate when the number of independent clusters is large, whereas a jackknife or another small-sample corrected variance estimator is preferable when K is small.

R Implementation for GEE

The `geepack` package can be used to fit generalized estimating equations through `geeglm()`.

```
library(geepack)

fit <- geeglm(
  y ~ x1 + x2,
  id = cluster_id,
  data = dat,
  family = binomial(),
  corstr = "exchangeable",
  std.err = "fij"
)
```

The main standard-error options are

`san.se`	ordinary sandwich standard errors,
`j1s`	one-step jackknife standard errors,
`jack`	approximate jackknife standard errors,
`fij`	fully iterated jackknife standard errors.

For a small number of clusters, `std.err = "fij"` is generally preferred.

Small-sample corrected inference can also be performed using the `clubSandwich` package:

```
library(clubSandwich)

coef_test(
  fit,
  vcov = "CR2",
  cluster = dat$cluster_id,
  test = "Satterthwaite"
)
```

Thus, a practical choice is

`geepack::geeglm()` for GEE fitting, and `clubSandwich::coef_test()` for small-sample corrected inference.